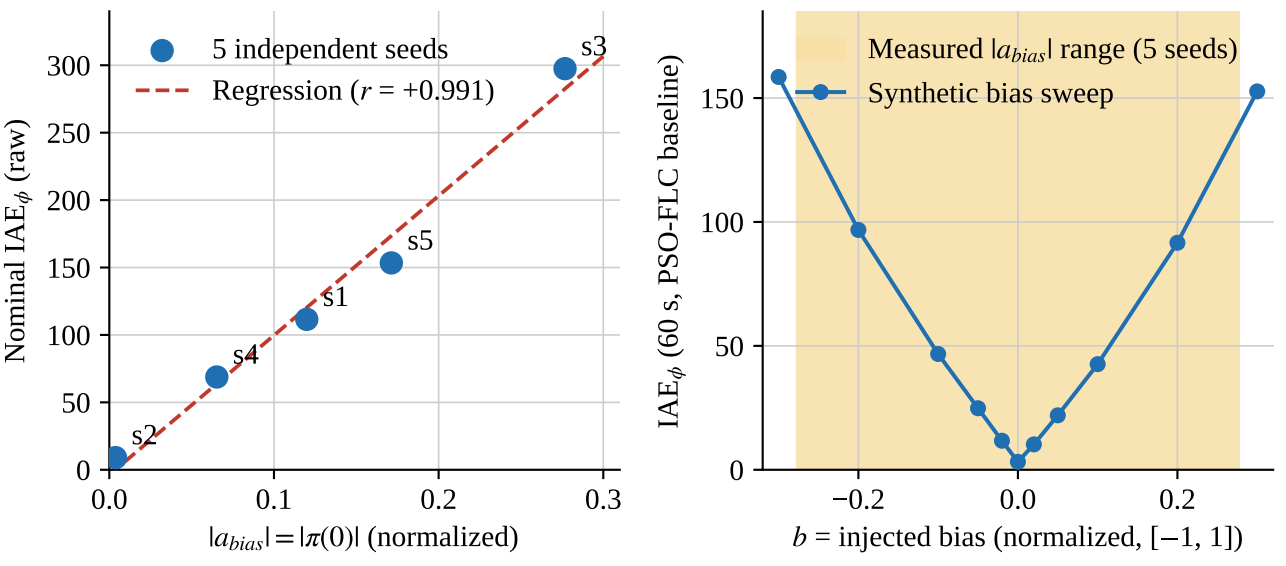
5 independent seeds
Regression ($r$ = +0.991)
s1
s2
s3
s4
s5
Nominal IAE$_\phi$ (raw)
$|a_{bias}| = |\pi(0)|$ (normalized)
0
50
100
150
200
250
300
0.0
0.1
0.2
0.3
Measured $|a_{bias}|$ range (5 seeds)
Synthetic bias sweep
IAE$_\phi$ (60 s, PSO-FLC baseline)
$b$ = injected bias (normalized, [−1, 1])
−0.2
0.0
0.2